Cross-Evaluation: Specialist Model Generalization (Dice Score)

Specialist model (trained on structure) \ Evaluation structure	ER (SERCA)	ER (SEC61β)	Golgi	Mitochondria	Adherens junctions	Actin bundles	Actomyosin bundles	Desmosomes	Gap junctions	Endosomes	Tight junctions	Nuclear pores	Nuclear speckles
ER (SERCA)	0.79 ±0.03	0.82 ±0.03	0.39 ±0.11	0.79 ±0.02	0.64 ±0.02	0.50 ±0.09	0.56 ±0.08	0.06 ±0.03	0.27 ±0.05	0.18 ±0.04	0.44 ±0.05	0.48 ±0.08	0.47 ±0.04
ER (SEC61β)	0.69 ±0.05	0.83 ±0.03	0.20 ±0.06	0.76 ±0.03	0.58 ±0.06	0.43 ±0.12	0.56 ±0.08	0.02 ±0.01	0.27 ±0.06	0.09 ±0.04	0.24 ±0.07	0.40 ±0.12	0.34 ±0.06
Golgi	0.50 ±0.02	0.54 ±0.02	0.80 ±0.04	0.75 ±0.02	0.51 ±0.03	0.37 ±0.11	0.44 ±0.06	0.37 ±0.08	0.24 ±0.08	0.19 ±0.04	0.30 ±0.07	0.38 ±0.07	0.47 ±0.07
Mitochondria	0.66 ±0.02	0.71 ±0.03	0.63 ±0.10	0.91 ±0.02	0.72 ±0.02	0.57 ±0.13	0.59 ±0.05	0.16 ±0.06	0.31 ±0.06	0.23 ±0.04	0.43 ±0.05	0.51 ±0.06	0.54 ±0.03
Adherens junctions	0.59 ±0.03	0.61 ±0.05	0.66 ±0.07	0.81 ±0.02	0.88 ±0.01	0.68 ±0.03	0.63 ±0.02	0.22 ±0.05	0.27 ±0.03	0.26 ±0.04	0.52 ±0.02	0.56 ±0.05	0.60 ±0.01
Actin bundles	0.64 ±0.02	0.62 ±0.04	0.57 ±0.10	0.82 ±0.01	0.70 ±0.02	0.67 ±0.11	0.67 ±0.06	0.10 ±0.05	0.32 ±0.06	0.19 ±0.04	0.57 ±0.05	0.51 ±0.08	0.61 ±0.04
Actomyosin bundles	0.64 ±0.02	0.65 ±0.03	0.60 ±0.09	0.76 ±0.02	0.68 ±0.02	0.67 ±0.07	0.72 ±0.07	0.15 ±0.06	0.32 ±0.06	0.27 ±0.04	0.67 ±0.04	0.56 ±0.08	0.66 ±0.02
Desmosomes	0.06 ±0.03	0.08 ±0.03	0.41 ±0.13	0.23 ±0.04	0.35 ±0.05	0.27 ±0.08	0.39 ±0.06	0.69 ±0.05	0.68 ±0.03	0.59 ±0.08	0.75 ±0.02	0.37 ±0.04	0.62 ±0.02
Gap junctions	0.02 ±0.01	0.03 ±0.01	0.32 ±0.14	0.25 ±0.07	0.38 ±0.10	0.19 ±0.11	0.40 ±0.09	0.27 ±0.13	0.73 ±0.03	0.60 ±0.20	0.81 ±0.02	0.60 ±0.12	0.72 ±0.07
Endosomes	0.12 ±0.03	0.17 ±0.03	0.56 ±0.16	0.42 ±0.05	0.57 ±0.04	0.54 ±0.08	0.56 ±0.05	0.68 ±0.06	0.67 ±0.04	0.67 ±0.07	0.80 ±0.02	0.74 ±0.03	0.73 ±0.02
Tight junctions	0.06 ±0.02	0.13 ±0.05	0.54 ±0.22	0.54 ±0.06	0.61 ±0.06	0.46 ±0.14	0.61 ±0.03	0.67 ±0.06	0.65 ±0.03	0.61 ±0.18	0.78 ±0.02	0.69 ±0.06	0.74 ±0.01
Nuclear pores	0.44 ±0.05	0.47 ±0.06	0.52 ±0.13	0.59 ±0.04	0.68 ±0.02	0.67 ±0.06	0.71 ±0.03	0.45 ±0.11	0.55 ±0.06	0.46 ±0.05	0.77 ±0.03	0.79 ±0.04	0.78 ±0.02
Nuclear speckles	0.35 ±0.05	0.49 ±0.05	0.44 ±0.18	0.67 ±0.04	0.68 ±0.03	0.61 ±0.07	0.73 ±0.06	0.73 ±0.06	0.52 ±0.05	0.56 ±0.08	0.79 ±0.03	0.71 ±0.04	0.78 ±0.02

Dice Score: 1.0, 0.8, 0.6, 0.4, 0.2, 0.0